

Isaiah Ruz

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EDUCATION

2024 - 2026	M.S. Physics at California State University, Fresno	(Degree GPA: 3.79/4.0)
2020 - 2022	B.A. Philosophy at University of California, Santa Cruz	(GPA: 3.47/4.0)
2013 - 2017	A.S. Physics at Cuesta College	(GPA: 3.28/4.0)

RESEARCH INTERESTS

I have a strong interest in applying and developing mathematical algebraic tools like cohomology in the study of topological defects and objects, especially in quantum field theory. I also have experience (discussed below) and interest in the study of entanglement entropy, especially in the areas of quantum information science, quantum field theory, and holography.

RESEARCH EXPERIENCE

California State University, Fresno (January 2024-May 2024)

Project: Calculating Generalized Uncertainty Principle Wavefunction Spread

Supervisor: Douglas Singleton and Michael Bishop

Explored a possible experimental setup to verify Kempf, Mann, and Mangano's Generalized Uncertainty Approach (GUP) to Quantum Gravity. This involved modifying the position operator in quantum mechanics to obtain a new uncertainty relation with a minimal length scale. Specifically, we calculated the spread of the width of the wavefunction of molecules in ordinary quantum mechanics and compared it to the calculated width in the GUP regime.

California State University, Fresno (May 2024-May 2025)

Project: Electromagnetic Field Momentum in Theoretical Magnetic Monopole-like Models

Supervisor: Douglas Singleton

We verified the Center-of-Energy Theorem by calculating the hidden momentum and field momentum of stationary theoretical magnetic monopole-like models (Dirac, Wu-Yang, Banderet), designing a methodology that could be used to examine other topological objects (e.g. Nieleisen-Olesen Vortex) to determine their physicality. My Master's thesis was built upon this research, where I did a deeper exploration of the gauge and fiber bundle theory behind the Wu-Yang monopole and provided the method from Theodore Frankel's "The Geometry of Physics" to prove the Dirac quantization condition that calculates the Chern number from the Chern class.

California State University, Fresno (August 2024-Present)

Project: An entanglement-entropy view of Unruh Temperature and a quantum optics experimental setup to calculate it.

Supervisor: Gerardo Muñoz

Extended the accelerated observer coordinates to all of spacetime to verify the interpretation of the Unruh Effect as a thermal state that arises from entanglement entropy. Determined a possible quantum optics experimental setup involving mechanical oscillators to measure its Unruh temperature, with the predicted Unruh temperature calculated to being within the measurable range for typical quantum optics experiments. This project is now focused on determining the Bekenstein-Hawking entropy of these two uncoupled mechanical oscillators after applying a squeeze operator on them.

PUBLICATIONS AND PREPRINTS

- Isaiah Ruz and Douglas Singleton. “Electromagnetic field momentum in theoretical magnetic monopole-like models”. In: *American Journal of Physics* 93.6 (June 2025), pp. 467–476. ISSN: 1943-2909. DOI: [10.1119/5.0244726](https://doi.org/10.1119/5.0244726). URL: <http://dx.doi.org/10.1119/5.0244726>.
- Isaiah Ruz and Gerardo Muñoz. *Measuring Unruh temperatures*. 2026. DOI: [10.2139/ssrn.6276070](https://ssrn.com/abstract=6276070). URL: <https://ssrn.com/abstract=6276070>.
- Isaiah Ruz. “The Application of Hidden Momentum and Gauge Theory to Magnetic Monopole-Like Models”. MS Thesis. California State University, Fresno, 2026.

CONFERENCE TALKS

Presented my talk “Hidden Momentum and Field momentum in the Wu-Yang Monopole Model” at APS Far West Section 2025

SCHOLARSHIPS AND AWARDS

Jon R. Dews Physics Research Scholarship	May 2025
Harry C. Burbidge Scholarship in Physics	May 2025
Fresno State Graduate Equity Fellowship	Dec 2025